5.5 Multi-head attention

- in practice, given the same set of queries, keys, and values, we may want our model to combine knowledge from different behaviors of the same attention mechanism, such as capturing dependencies of various ranges (e.g., shorter-range vs. longer-range) within a sequence
- thus, it may be beneficial to allow our attention mechanism to jointly use different representation subspaces of queries, keys, and values

5.5 Multi-head attention

- to this end, instead of performing a single attention pooling, queries, keys, and values can be transformed with *h* independently learned linear projections
- then, these *h* projected queries, keys, and values are fed into attention pooling in parallel
- in the end, the *h* attention pooling outputs are concatenated and transformed with another learned linear projection, to produce the final output
- this design is called *multi-head attention*, where each of the *h* attention pooling outputs is a *head*
- using fully-connected layers to perform learnable linear transformations, Figure 6 describes multi-head attention

5.5 Multi-head attention

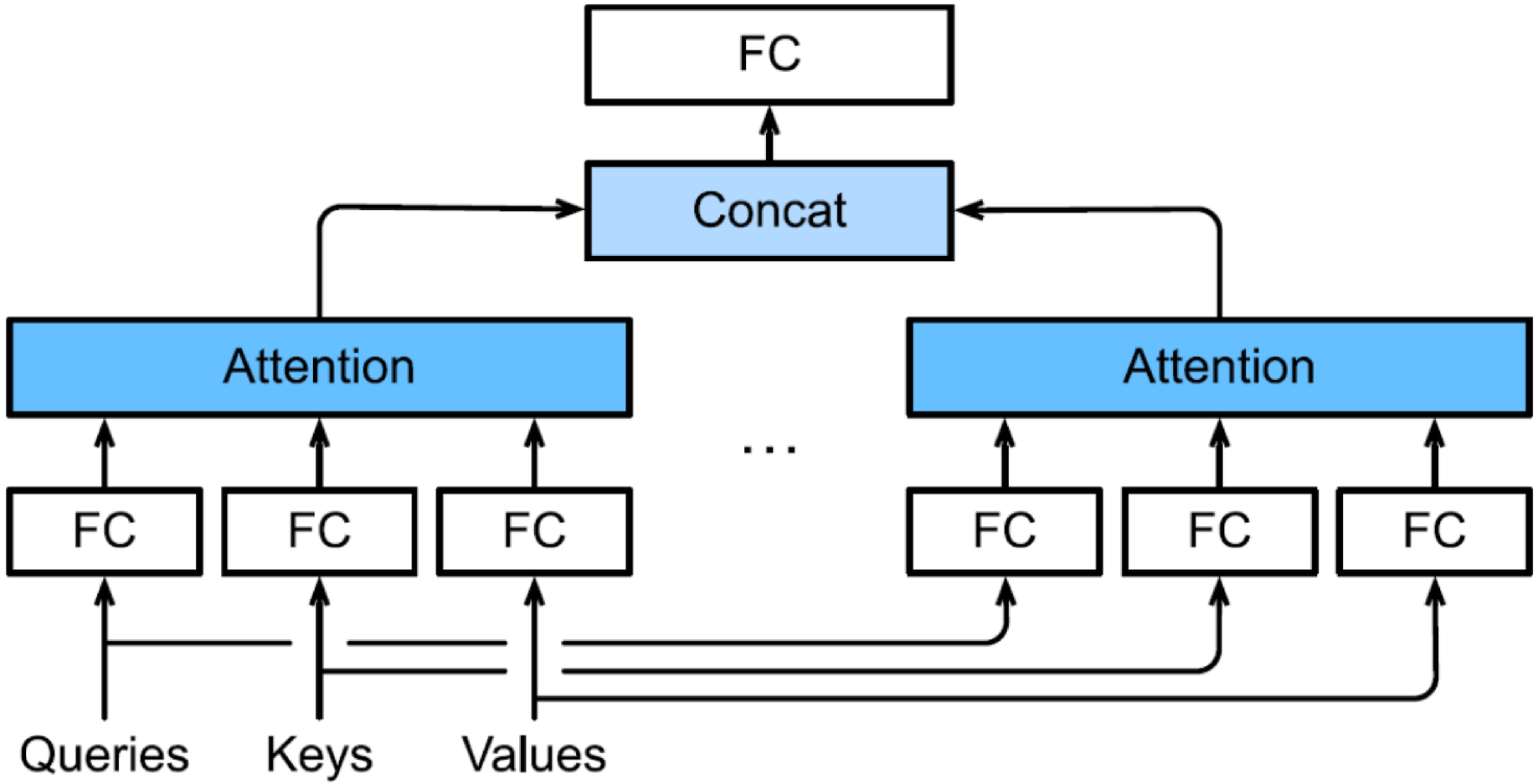


Figure 6: Multi-head attention, where multiple heads are concatenated, then linearly transformed.

5.5 Multi-head attention

- let us formalize this model mathematically; given a query $\boldsymbol{q} \in \mathbb{R}^{d_q}$, a key $\boldsymbol{k} \in \mathbb{R}^{d_k}$, and a value $\boldsymbol{v} \in \mathbb{R}^{d_v}$, each attention head $\boldsymbol{h}_i$ ($i = 1, \ldots, h$) is computed as:

$$\boldsymbol{h}_i = f(\boldsymbol{W}_i^{(q)}\boldsymbol{q}, \boldsymbol{W}_i^{(k)}\boldsymbol{k}, \boldsymbol{W}_i^{(v)}\boldsymbol{v}) \in \mathbb{R}^{p_v},$$

  where the parameters $\boldsymbol{W}_i^{(q)} \in \mathbb{R}^{p_q \times d_q}$, $\boldsymbol{W}_i^{(k)} \in \mathbb{R}^{p_k \times d_k}$, and $\boldsymbol{W}_i^{(v)} \in \mathbb{R}^{p_v \times d_v}$ are learnable, and f is attention pooling, such as additive attention or scaled dot-product attention, discussed in Section 5.3
- the multi-head attention output is another linear transformation via learnable parameters $\boldsymbol{W}_o \in \mathbb{R}^{p_o \times hp_v}$ of the concatenation of the h heads:

$$\boldsymbol{W}_o \begin{bmatrix} \boldsymbol{h}_1 \\ \vdots \\ \boldsymbol{h}_h \end{bmatrix} \in \mathbb{R}^{p_o}.$$

- based on this design, each head may attend to different parts of the input; more sophisticated functions than the simple weighted average can be expressed